

HPS Gaussian-Process Resonance Search

Analysis Note v4.9.12: Pointwise Expected-Limit Bands

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Initial band stage. This results-only note uses 50 background-only toys per mass (toy IDs 0–49). The outer expected band is therefore provisional and will be sharpened by the planned 100- and 300-toy stages. All bands are pointwise and conditional on frozen GP states.

1 Inputs and construction

The seven scopes are the exact final-sample curves used in the most recent Harvard results section: full 2015, full 2016, the optimized-support 2021 10% sample, their three pairs, and the all-three shared- ϵ^2 combination. The 2021 configuration uses the frozen 36–300 MeV GP support and factor-15 resolution-scaled length ceiling. No 2021 1% or 2016 10% comparison enters these results.

At each mass, a latent background vector is drawn from the conditioned mass-local GP prediction and then Poisson sampled. The same dataset draw is reused by every scope containing that dataset at a fixed mass and toy ID, so the standalone and combined comparisons are paired. GP hyperparameters are not refit inside toys. Each pseudoexperiment is evaluated with the same bounded, piecewise-asymptotic 90% CL_s construction as the observed v4.9.12 curve. For numerical stability, fixed-strength nuisance fits minimize a data-constant-centered Poisson deviance; the fitted likelihood, statistic, and root acceptance gates are unchanged from the parent solver.

The bands are empirical 2.5%, 16%, 50%, 84%, and 97.5% quantiles. Because different mass points are generated independently, these are pointwise expected-limit bands rather than a scan-wide ensemble. They do not establish unconditional coverage, toy-calibrated CL_s , a look-elsewhere correction, or global significance. Full-2016 and 2016-inclusive curves also retain the previously disclosed cross-process GP-state replay exception.

2 Results

Figure 1 is the direct Brazil-band analogue of the earlier v4.2/v4.5 combination display, now evaluated on the current final inputs. At this initial stage, local motion in the outer envelope should be read as finite-ensemble granularity rather than a stable tail feature.

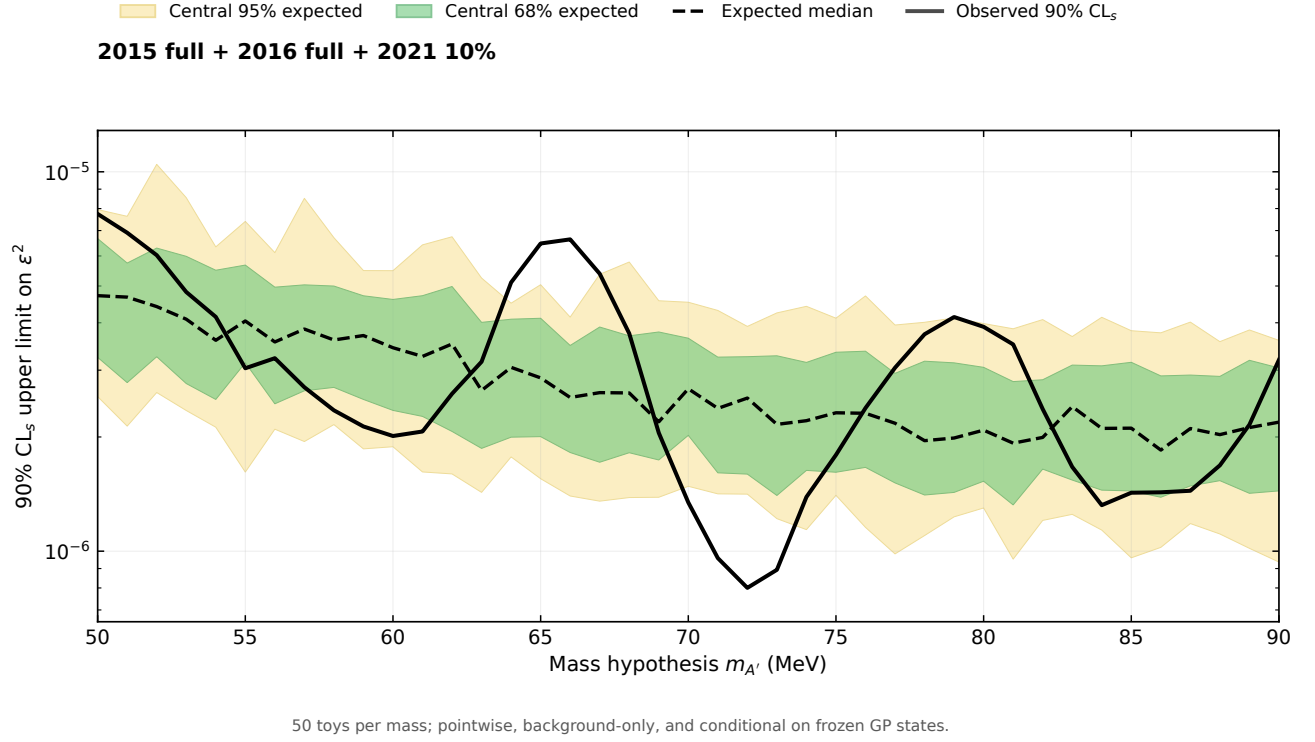


Figure 1: Observed and expected 90% CL_s upper limit for the all-three shared-coupling combination. Yellow and green show the central 95% and 68% pointwise background-only ranges; the dashed curve is the expected median. The legend is outside the data region so it does not obscure any curve.

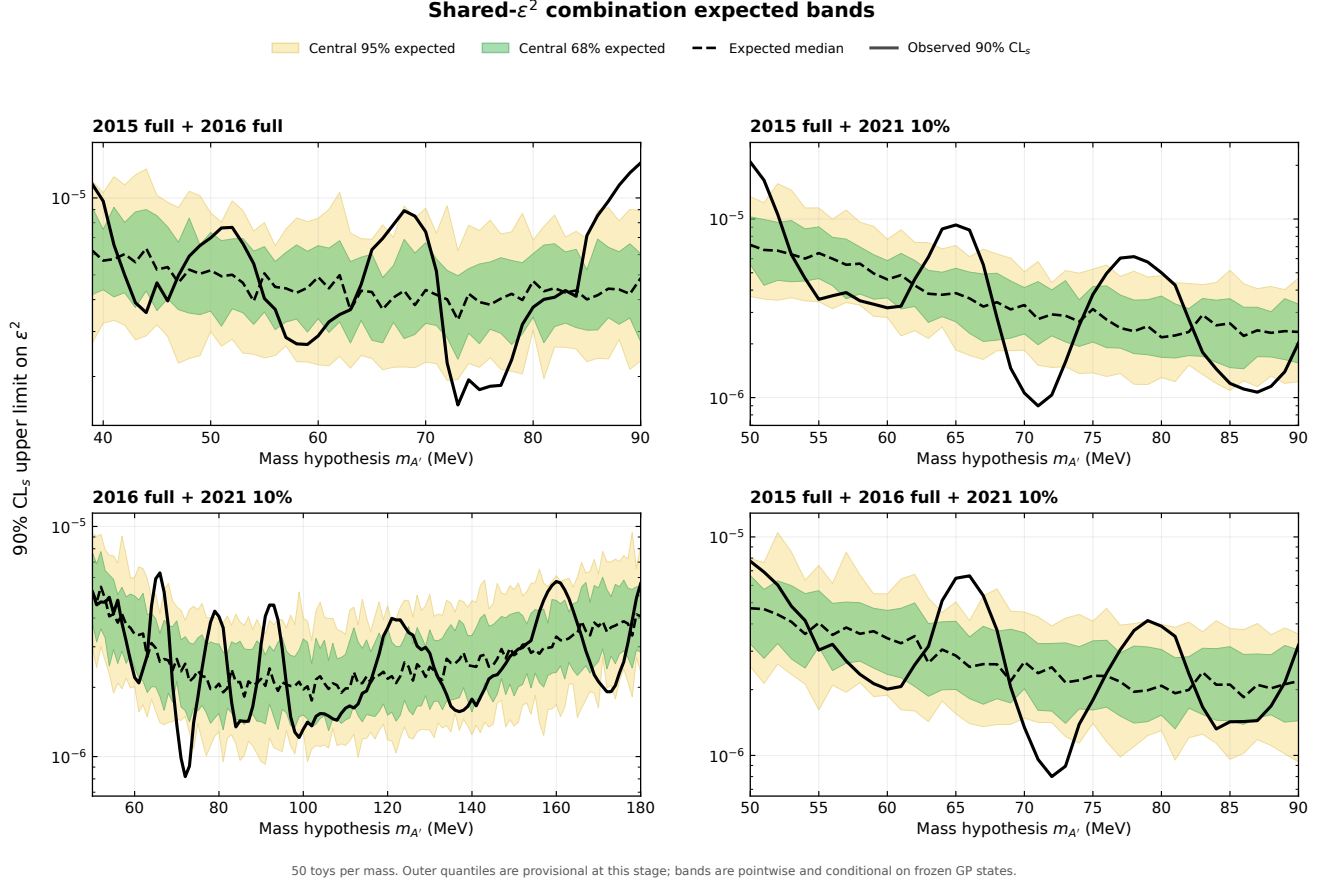


Figure 2: Pairwise and all-three shared- ϵ^2 limits. Each combination has its own axis, preventing bands or observed curves from covering another result. All four panels use paired constituent pseudo-observations.

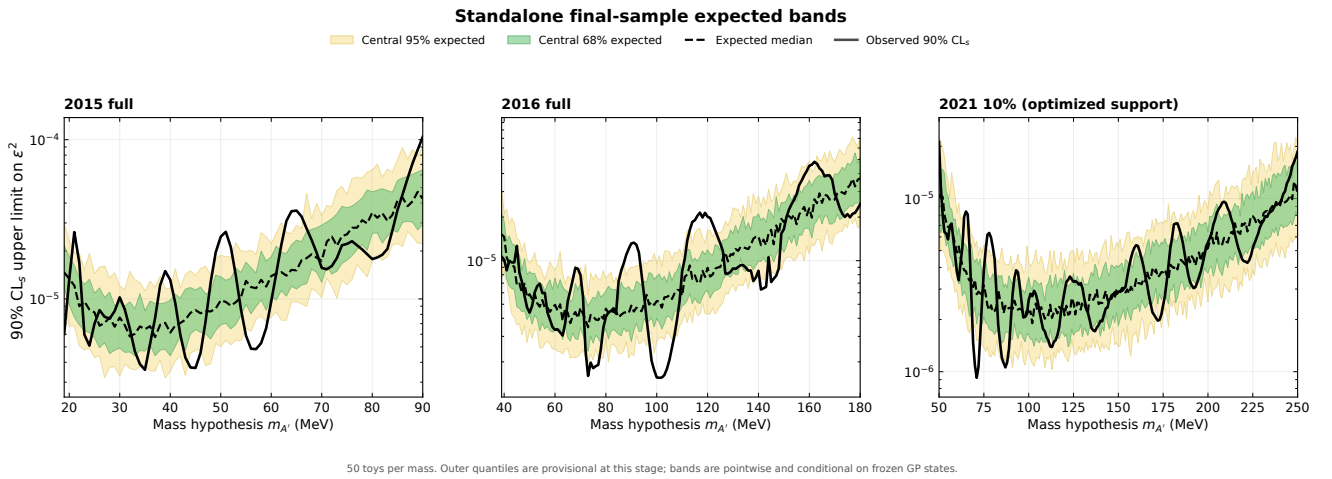


Figure 3: Standalone final-sample limits. The 2021 panel is the optimized native-10% configuration with 36–300 MeV GP support.

2.1 Compact numerical summary

Table 1 reports, for each scope, the mass at which the 50-toy expected median is smallest. These minima summarize the plotted curves; they are not optimization inputs and do not select the analysis card.

Scope	m [MeV]	Observed	-2σ	Median	$+2\sigma$
2015 full	37	7.582×10^{-6}	3.520×10^{-6}	5.757×10^{-6}	1.180×10^{-5}
2016 full	73	1.600×10^{-6}	1.996×10^{-6}	3.467×10^{-6}	5.820×10^{-6}
2021 10%	102	2.726×10^{-6}	1.119×10^{-6}	1.903×10^{-6}	4.215×10^{-6}
2015 + 2016	73	1.548×10^{-6}	1.948×10^{-6}	3.334×10^{-6}	6.212×10^{-6}
2015 + 2021	80	5.005×10^{-6}	1.294×10^{-6}	2.177×10^{-6}	4.815×10^{-6}
2016 + 2021	102	1.360×10^{-6}	1.078×10^{-6}	1.752×10^{-6}	3.194×10^{-6}
All three	86	1.429×10^{-6}	1.023×10^{-6}	1.847×10^{-6}	3.763×10^{-6}

Table 1: Observed and conditional expected ϵ^2 limits at each scope’s minimum expected-median mass. The outer columns are empirical central-95% endpoints from 50 toys per mass.

Across all 680 scope–mass coordinates, all 34,000 toy limits were finite and closed at $CL_s=0.10$ within the frozen numerical tolerance. No latent GP draw used the clipping fallback. The exact per-toy limits and pseudo-observation hashes are retained in machine-readable ledgers.

3 Continuation

The ensemble is cumulative. Re-running the driver with `--target-toys 100` appends IDs 50–99; using `--target-toys 300` then appends IDs 100–299. Atomic per-mass checkpoints make either stage resumable, and the contract hash prevents a changed card, GP-state ledger, observed curve, solver, or protocol from being silently mixed into the existing toys.